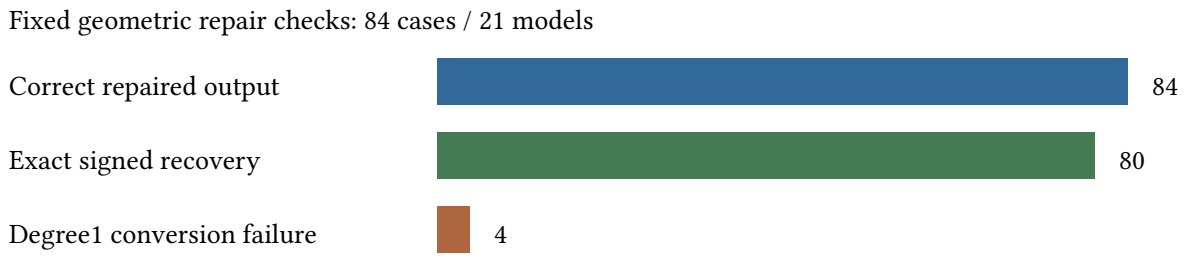

Overall frozen all84 gate: FAIL (exit101). Evidence: run.stdout.

Successful geometric repair and explicit evaluation masks

The fixed software checks cover 84 valid geometric repair cases across21 retained models. All84 cleared mixed additions initially return zero, then torsion-translation repair succeeds and agrees with direct signed Cantor addition. All84 selected eager/pruned pairs and evaluation/selection-mask checks agree. Signed recovery equals the expected signed divisor in80 cases; four p739 cases return general_conversion_inapplicable. The prospective requirement was all84: the overall check exits101, preserved without dropping failures.

Construction and scope

Take each pinned exceptional identity witness T with two zero Kummer coordinates. T is a valid rational2-torsion divisor. From retained Q form the adjacent pair(Q,Q+T). The mixed result is2Q+T. Verify2T=0, validity, rG=rQ=0, and the explicit word bounds before checks. Recover a resulting point R using the original non2-torsion signed G and the pair(kappa(R),kappa(R+G)). This external anchor is essential: the CCS paper excludes2-torsion anchors. These fixtures qualify repaired arithmetic and signed-recovery composition, not a collector with a torsion sign anchor.

Sources: Chung-Costello-Smith, sections2 and6.4, https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/recovery.pdf; pinned historical N19 main_labelled.rs and walk_impl_uncached.rs. Existing84 ordinary/forced-fallback checks are separate and unchanged.

Telemetry candidate

The additive candidate carries evaluated[3] and selected[3] alongside repair roots. Eager evaluation is[true,true,true]. Pruned evaluation equals selection: [true,true,false] for branch0 and[false,true,true] for branch1. Unexecuted now differs from evaluated-without-recorded-repair. Option output still distinguishes failure from success. Fixed cases cover52 branch0 and32 branch1 decisions. Native per-case walk counter reductions are24M,36A,4I, not a timing or complete collector cost claim.

Retained recovery failure

All four failures yield the same degree1 divisor on p739: u=[362,1,0],v=[303,0], for exceptional indices8,12,16,20. Both versions fail identically. Generic-coordinate conversion rejects a zero first homogeneous coordinate, although the forward map explicitly represents degree1 as(0:1:x:x^2). This provides a concrete next construction: decode that boundary chart, enumerate its at most two curve lifts and use signed anchor addition to choose uniquely. Identity, zero vector, nonsquare and ambiguous anchors must remain explicit.

Reproduction

PLAN.md freezes selection and assertions before execution. Source/fixture hashes are in pre-execution-manifest.json; receipt.json retains the bounded locked/offline command and exit101. No new scientific solves, cloud operations, runtime updates or publication occurred. Candidate patch remains private and requires independent review.